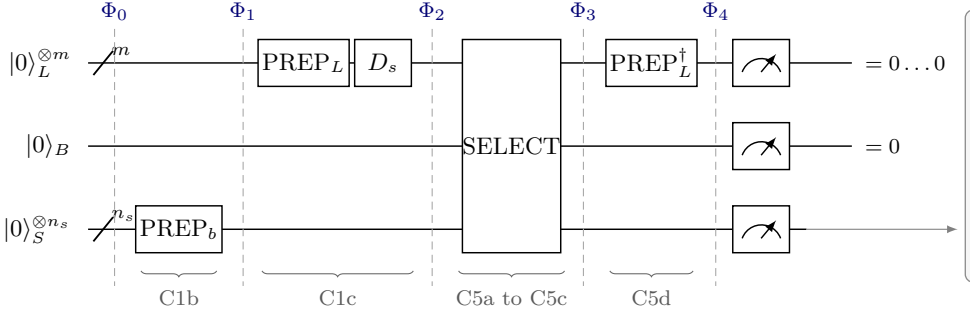
Full computation 3: segment register, QSVT polynomial filter. (a) LCU of walk powers, as built in `QSVT.build_circuit`

Classical
$A = sI - C$, $\mathbf{b} = \delta\mathbf{1}$
$X = \dfrac{2A - (d_{lo} + d_{hi})I}{d_{hi} - d_{lo}}$
design $p = \sum_{k \le d} c_k T_k$
(line comb, $d = 40$)
PREP angles, signs



Classical
keep shots with
$L = 0$, $B = 0$
($\approx M \ln M/P$ shots)
$\hat{x}_i$, threshold
chain segments (`get_tracks`)

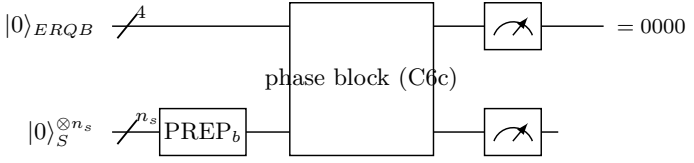
$\Phi_1 = |0\rangle_L|0\rangle_B|b\rangle$, $|b\rangle = N^{-1/2}\sum_{i<N}|i\rangle$ $\quad \Phi_2 = \sum_k \mathrm{sgn}(c_k)\sqrt{|c_k|/\|c\|_1}\,|k\rangle_L|0\rangle_B|b\rangle$ $\quad \Phi_3 = \sum_k \mathrm{sgn}(c_k)\sqrt{|c_k|/\|c\|_1}\,|k\rangle_L\big(|0\rangle_B T_k(X) - |1\rangle_B S_k\big)|b\rangle$

$(\langle 0|_L\langle 0|_B)\Phi_4 = p(X)|b\rangle/\|c\|_1$, $P = \|p(X)b\|^2/\|c\|_1^2$; item i with probability $[p(X)b]_i^2/\|p(X)b\|^2$ (7.11)

Worked event: $n_s = 5$, $m = 6$, 12 qubits; $p(2.382) = 0.348$, $p(4.618) = 0.183$, $p(4) = -0.0016$; $P = 0.176$; readout 0.313 (end), 0.390 (middle), 0.0017 (false). Width $n_s + 1 + m$; $2^m - 1$ walk steps per shot as built.

(b) The same filter by a phase sequence (QSVT proper; `QSVT.solve_phase_sequence`, simulated as linear algebra, not built)

Classical
parity split $q = p/\max|p| = \alpha_e q_e + \alpha_o q_o$
phases Φ_e, Φ_o
(`find_phase_factors`)



Post-selected: $q(X)|b\rangle/(\alpha_e + \alpha_o)$, $P = \|q(X)b\|^2/(\alpha_e + \alpha_o)^2$.
Width $n_s + 4$, $4d = 160$ calls of U per shot at $d = 40$; P larger than the LCU's by $(\|c\|_1/\max|p|)^2$, a factor 2.2 for the production comb (7.4).
Not run on the worked event.